



“**BRAILLE IS KNOWLEDGE; KNOWLEDGE IS POWER.**”

Louis Braille, French inventor of Braille writing

A simple invention by a blind French boy to help him read, Braille has become a window into the world of books for millions of visually impaired people.

DURING 1823–24, SCIENTISTS WERE STUDYING THE NIGHT

sky as well as the history of Earth. Bavarian astronomer **Franz von Gruithuisen** (1774–1852) realized craters on the Moon were formed by past meteorite impacts. Another German astronomer, **Heinrich Olbers** (1758–1840), asked why the night sky is dark. Surely, if there is an infinite number of stars, then it should be possible to see a star in every direction and, as a result, the night sky should be bright. Olbers was not the first scientist to ask this question, but it has become known as **Olbers' Paradox**. Today, this paradox is known to be the result of space expanding, which diminishes the apparent brightness of distant stars in many directions, causing the sky to appear dark.

British naturalist **William Buckland** made two momentous contributions to the field of geology. The first was his **discovery**, in a cave on the coast of Wales, UK, of the **first fossilized human remains ever found**. Buckland wrongly identified them as being those of a Roman woman. Carbon-dating (see 1955) has since confirmed they are, in fact, of a 33,000-year-old man. Buckland's second contribution, in 1824, was when he gave the first scientific

description of a dinosaur (although the term was not coined until 1842)—he identified some fossils as a giant extinct lizard called **megalosaurus**.

French mathematician **Joseph Fourier**, calculated that Earth is too far from the Sun to be warmed to the temperature it is by solar radiation alone. He



JOSEPH FOURIER
(1768–1830)

Joseph Fourier was a brilliant mathematician who went with Napoleon to Egypt in 1798 to decode hieroglyphs. He studied heat transfer and identified the greenhouse effect. His studies of waves led to Fourier analysis, the mathematical analysis of wave forms, now used in everything from touch screens to understanding brain function.

Megalosaurus bones

Drawings of megalosaurus bones from William Buckland's 1824 paper, which contained the first scientific description of a dinosaur.

suggested that heat is trapped by Earth's atmosphere. This was the first identification of what later became known as **the greenhouse effect**.

Hungarian mathematician **János Bolyai** (1802–60) pioneered a new form of geometry—**non-Euclidean geometry**.

It breaks away from Euclid's definition of parallel lines on a flat, two-dimensional surface (see 400–335 BCE), and frees mathematicians to contemplate abstract multidimensional ideas, such as the **curved nature of space, time, and the Universe**, and parallel lines that can actually cross.

A blind 15-year-old French boy called **Louis Braille** (1809–52) invented the **six-dot code** later known as **Braille**. This writing system enables blind or partially sighted people to read and is now used in virtually every country.

Also in France, engineer **Nicolas Sadi Carnot** (1796–1832)



published *Reflections on the Motive Power of Fire and on Machines Fitted to Develop that Power*. This book contained the **first successful theory of heat engines**, which is now known as the **Carnot cycle**. All heat engines are inefficient because

they lose heat every time hot gases are released before the next cycle. The Carnot cycle shows the maximum theoretical efficiency for all engines. Carnot laid the foundations for the **science of thermodynamics** (see 1847–48).

700 THE NUMBER OF **DINOSAUR SPECIES** THAT HAVE BEEN IDENTIFIED

1823 The **Fresnel lens** is installed in Cordouan lighthouse in France

1823 William Buckland discovers **prehistoric human remains**

1823 Heinrich Olbers outlines **Olbers' paradox**

1823 János Bolyai invents **non-Euclidean geometry**

1823 British chemist Charles Macintosh invents **waterproof fabric**

1824 William Buckland makes **first description of a dinosaur**

1824 Nicolas Léonard Sadi Carnot establishes **thermodynamics**

1824 Joseph Fourier discovers the **greenhouse effect**

1824 Franz von Gruithuisen suggests **Moon craters** were formed by meteor impacts

1824 Louis Braille invents **Braille**